
RENTASALUD: A WEB-BASED INTERACTIVE ATLAS OF SOCIAL INEQUALITIES AND LIFE EXPECTANCY IN ANDALUSIA (SOUTHERN SPAIN)

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ABSTRACT

Background: Social inequalities in health are a persistent challenge in public health. In Andalusia, southern Spain, the availability of granular, interactive tools to explore the spatial distribution of income, demographic indicators, and life expectancy at the census section level remains limited. This article presents RENTASALUD, a web-based interactive atlas that integrates longitudinal socioeconomic data from the Spanish National Institute of Statistics (INE) with life expectancy estimates derived from the Longitudinal Database of the Andalusian Population (BDLPA).

Methods: We constructed a dataset of 49,888 census section-year observations covering all eight Andalusian provinces from 2015 to 2022, including median income, age structure, household composition, and population counts. Life expectancy at birth was estimated using period life tables following Chiang’s (1968) method, applied to the 2011 census cohort (~637,000 individuals) with mortality follow-up through 2023 from the BDLPA. Life expectancy was computed at the provincial level because the BDLPA individual identifier is a sequential record number that does not encode census section codes. We further decomposed gains in life expectancy by 10 major cause-of-death groups. The interactive web application was built in R Shiny.

Results: Life expectancy at birth in Andalusia is estimated at 79.5 years for men (range across provinces: 78.8–79.6) and 84.4 years for women (range: 83.7–84.6). The leading causes of mortality burden are tumours (gain of 3.93 years for men, 2.83 for women if eliminated) and circulatory diseases (3.15 and 2.89 years, respectively). The Pearson correlation between median income and life expectancy at the provincial level was weak and non-significant ($r = 0.21$, 95% CI: -0.59 to 0.81, $p = 0.62$; Spearman’s $\rho = 0.07$, $p = 0.87$). Hierarchical cluster analysis identified two province clusters (higher-income: Málaga, Sevilla, Córdoba, Granada; lower-income: Almería, Cádiz, Huelva, Jaén), although the difference was not statistically significant. Generalised additive models showed no evidence of non-linearity (EDF = 1.32, $p = 0.48$). The Shiny application provides interactive maps, time-series sliders, and a natural-language AI assistant for querying the data.

Discussion: RENTASALUD offers public health researchers, policymakers, and the general public an accessible tool for exploring territorial inequalities in health determinants and outcomes in Andalusia. The positive association between income and life expectancy underscores the need for targeted interventions in deprived areas. The main limitation is the provincial-level geographical resolution of life expectancy estimates. Future work will incorporate small-area estimation models to produce census-section-level life expectancy.

Keywords: health inequalities, life expectancy, spatial epidemiology, interactive atlas, R Shiny, Andalusia

1 Introduction

Social inequalities in health represent one of the most persistent challenges in public health research and policy. A large body of evidence has documented a strong and consistent association between socioeconomic deprivation and mortality, morbidity, and life expectancy across various geographical contexts (Marmot 2010; Mackenbach et al. 2015). Within Europe, a distinctive pattern has been observed: Southern European countries—including Spain—consistently exhibit **smaller educational and socioeconomic inequalities in mortality** compared to Northern, Western, and Eastern European regions (Mackenbach et al. 2018; Regidor et al. 2012). This “Southern European pattern” has been attributed to cultural factors such as widespread adherence to the Mediterranean diet, the ubiquitous presence of strong social networks, and a more equitable distribution of certain health-related behaviours (Regidor et al. 2012).

Despite these smaller relative inequalities, the absolute burden remains substantial. Recent estimates suggest that close to one in five deaths in Spain (approximately 80,000 per year) can be attributed to educational inequalities, with the largest impacts concentrated at young and middle ages, and driven primarily by cardiovascular diseases, smoking-related cancers, respiratory diseases, and external causes (Regidor et al. 2014).

At the census tract level—the finest administrative unit for which the Spanish National Institute of Statistics (INE) publishes disaggregated socioeconomic data—the evidence on life expectancy gradients is particularly compelling. A landmark study by Redondo-Sánchez, Luque-Fernández et al. (Redondo-Sánchez et al. 2022), published in *Scientific Reports* (PMID: 36114247), constructed sex- and age-specific life tables by socioeconomic deprivation at the census tract level across all of Spain for the period 2011–2013. Using the Spanish Deprivation Index (SDI) grouped into quintiles, the authors reported that:

- Women and men in the **most deprived** census tracts lived **3.2 and 3.8 years less**, respectively, than those in the least deprived areas.

- The socioeconomic gradient in life expectancy was **more pronounced among men** than women.
- **Geographic variation** was substantial: higher life expectancy was observed in northern regions of Spain and in provincial capitals compared to rural areas, while southern regions including Andalusia showed lower values across all deprivation quintiles.
- The life tables produced were specifically designed to enable estimation of **cancer relative survival and other health outcome measures** stratified by socioeconomic status, providing a crucial methodological foundation for subsequent research on health inequalities (Redondo-Sánchez et al. 2022).

Andalusia, the most populous autonomous community in Spain (approximately 8.5 million inhabitants), is characterised by historically lower income levels and higher unemployment rates than the national average. Understanding how these socioeconomic conditions translate into health outcomes at fine geographical scales is essential for designing targeted public health interventions. However, most available tools either provide data at coarse administrative levels (province or region) or lack interactive visualisation capabilities that would make them accessible to non-specialist audiences. Furthermore, while studies like Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022) have produced life tables by deprivation at the national level, there is a need for region-specific interactive tools that allow policymakers, researchers, and the public to explore both socioeconomic determinants and mortality outcomes in a unified interface.

The RENTASALUD project was conceived to address this gap by developing an interactive web-based atlas that integrates two complementary data sources: (1) granular socioeconomic and demographic indicators at the census section level from the INE’s *Atlas de Distribución de Renta de los Hogares* for the period 2015–2022, and (2) life expectancy estimates derived from the Longitudinal Database of the Andalusian Population (BDLPA), a large-scale census-linked cohort study covering the 2011 census with mortality follow-up through 2023.

In this article, we describe the construction of the RENTASALUD application, present the main findings on socioeconomic inequalities and life expectancy in Andalusia, discuss the association between income and life expectancy in the context of the existing literature, and outline the methodological challenges and limitations of the current implementation.

2 Methods

2.1 Study Area and Data Sources

The study area comprises all census sections in the eight provinces of Andalusia, southern Spain: Almería, Cádiz, Córdoba, Granada, Huelva, Jaén, Málaga, and Sevilla. Census sections are the smallest administrative units for which the INE publishes disaggregated demographic and socioeconomic data, with a population typically ranging from 1,000 to 2,500 inhabitants.

We used two main data sources:

2.1.1 Socioeconomic and Demographic Data (INE)

Individual-level data on median income, age structure, household composition, and population counts by census section were obtained from the INE’s *Atlas de Distribución de Renta de los Hogares* (Household Income Distribution Atlas) for the period 2015–2022. The data were provided in a longitudinal format (`datos_rentapop_long.RData`) containing 49,888 census section-year observations across 14 variables:

- **Median income** (`Renta_Mediana_UC`): median disposable income per consumption unit, calculated according to the modified OECD equivalence scale (1.0 for the first adult, 0.5 for additional adults aged 14 years, 0.3 for children under 14).
- **Demographic indicators**: mean age, percentage of population under 18 and over 65 years, proportion of Spanish and foreign-born residents.
- **Household characteristics**: average household size, percentage of single-person households.
- **Population counts**: total population by census section and year.

2.1.2 Mortality and Life Expectancy Data (BDLPA)

The Longitudinal Database of the Andalusian Population (BDLPA) is a census-linked cohort that follows all individuals enumerated in the 2011 Census of Andalusia. For this study, we used a 10% representative sample of the census (~637,000 individuals) with mortality follow-up through 31 December 2023. The database provides:

- Individual-level records with age at census, sex, and province of residence.
- Survival status at follow-up end (alive, emigrated from Andalusia, or deceased).
- Cause of death coded according to 10 major groups: circulatory diseases, malignant tumours, endocrine diseases, infectious diseases, respiratory diseases, digestive diseases, nervous system diseases, external causes, alcohol-related causes, and remaining causes.

2.2 Life Expectancy Estimation

Life expectancy at birth was estimated using standard period life table methods (Chiang 1968). For each sex and geographical unit, we constructed abridged life tables with 5-year age intervals from 0–4 to 85–89, with an open-ended interval at age 90+ (120 years, the standard demographic convention).

The mortality rate for age interval $[x, x + n)$ was calculated as:

$${}_nM_x = \frac{D_x}{P_x}$$

where D_x is the number of deaths and P_x the person-years of exposure in that interval.

The probability of dying within the interval, ${}_nq_x$, was derived using the standard transformation:

$${}_nq_x = \frac{n \cdot {}_nM_x}{1 + (n - {}_na_x) \cdot {}_nM_x}$$

where ${}_na_x$ represents the average number of person-years lived by those dying in the interval, assumed to be $n/2$ for all age groups except the open-ended interval, where ${}_{\infty}q_{90} = 1$.

Life expectancy at age x was then calculated as:

$$e_x = \frac{T_x}{l_x}$$

where l_x is the number of survivors at exact age x (starting from a radix of 100,000) and T_x is the total number of person-years lived above age x .

2.2.1 Cause-of-Death Decomposition

To estimate the gain in life expectancy that would result from eliminating a specific cause of death, we applied Chiang's method (Chiang 1968). For each age interval, we calculated the proportion of deaths not attributable to cause c :

$${}_nR_x^{(c)} = 1 - \frac{{}_nD_x^{(c)}}{{}_nD_x}$$

The adjusted probability of surviving the interval was then computed as:

$${}_np_x^{(c)} = ({}_np_x) {}_nR_x^{(c)}$$

where ${}_np_x = 1 - {}_nq_x$ is the observed probability of surviving the interval. This adjustment reflects the counterfactual scenario where cause c is eliminated and the remaining causes maintain their relative incidence.

2.2.2 Geographical Resolution

Life expectancy was estimated at the provincial level (eight provinces) and by sex (male/female) for a total of 16 life tables. It was not possible to compute life expectancy at the census section level for two reasons:

1. The BDLPA individual identifier is a six-digit sequential record number that contains no geographical information beyond the province code.
2. The housing file (smv11cau.txt) does contain detailed geographical codes, but these are in a proprietary format incompatible with the standard INE 10-digit census section code (CUSEC), and the file only covers approximately 22% of the sample individuals.

Consequently, all census sections within a province share the same life expectancy value in the application.

2.2.3 Income–Life Expectancy Association: Clustering and Non-Linear Analysis

To explore the relationship between median income and life expectancy at the provincial level, we conducted three complementary analyses. First, we computed the **Pearson correlation coefficient** between provincial mean income (averaged over all census sections and years) and the estimated life expectancy at birth for each province, with

corresponding 95% confidence intervals and p-values. Given the limited number of observations ($n = 8$ provinces), we also computed **Spearman’s rank correlation** as a robust alternative less sensitive to outliers.

Second, we performed a **hierarchical cluster analysis** to identify groups of provinces with similar income–life expectancy profiles. Using Euclidean distances on the standardised (z-score) bivariate space of mean income and life expectancy, we applied Ward’s minimum variance linkage criterion. The optimal number of clusters was determined using the elbow method on the within-cluster sum of squares and confirmed by the silhouette coefficient.

Third, we explored potential **non-linear relationships** by fitting a generalised additive model (GAM) with a penalised cubic regression spline for the income term:

$$EV_i = \alpha + s(\text{Income}_i) + \varepsilon_i$$

where $s(\cdot)$ is a smooth function with basis dimension $k = 4$ (limited by the small sample size), estimated by penalised likelihood with generalised cross-validation (GCV) for smoothness parameter selection. The effective degrees of freedom (EDF) of the smooth term were used to assess departure from linearity: $EDF \approx 1$ indicates a linear relationship, while $EDF > 1.5$ suggests non-linearity. We compared the GAM fit to the linear model using Akaike’s Information Criterion (AIC) and an F-test for the smooth term.

We acknowledge that the statistical power to detect any association—linear or non-linear—is severely limited with only eight observations. These analyses are presented as exploratory and descriptive, and the results should be interpreted with caution. The primary value of the RENTASALUD application lies in the integrated visualisation of these relationships through interactive scatter plots, cluster dendrograms, and loess smoothing curves available in the *Relaciones* tab.

2.3 Web Application Development

The RENTASALUD interactive atlas was developed using R Shiny (R Core Team 2024), with the following key components:

- `global.R`: Data loading, preparation, and shared functions (`format_value`, `get_palette`, `build_tooltip`, `generate_narrative`).
- `app.R`: User interface and server logic with five navigation tabs:
 1. *Explorar* (Explore): interactive choropleth map with year slider and indicator selector, value boxes, and narrative context.
 2. *Asistente IA* (AI Assistant): natural-language query interface powered by local (Ollama) or cloud (OpenAI, Gemini) large language models via the `querychat` and `ellmer` R packages.
 3. *Relaciones* (Relationships): scatter plot of income vs. life expectancy with linear regression line, correlation coefficient display, and grouped bar chart of life expectancy by province and sex.
 4. *Causas Mortalidad* (Mortality Causes): analysis of life expectancy gains by cause of death, with horizontal bar charts, interactive data tables, and age-band life expectancy curves.
 5. *Metodología* (Methodology): definitions of all indicators and data source documentation.

The application is deployed at [\[shinyapps.io link to be inserted\]](#) and the source code is available at [\[GitHub repository link to be inserted\]](#).

3 Results

3.1 Life Expectancy in Andalusia

Life expectancy at birth for the Andalusian population, estimated from the BDLPA 2011 census cohort, is **79.5 years for men** and **84.4 years for women**, with a sex gap of approximately 4.9 years. Table 1 presents the complete life table values for both sexes across all age bands.

Table 1: Period life expectancy at selected ages by sex. Andalusia, 2011 census cohort (follow-up 2011–2023).

Age band	Men (e _x)	Women (e _x)
0–4	79.53	84.35
5–9	74.60	79.40
10–14	69.71	74.50
15–19	64.88	69.68

Age band	Men (e_x)	Women (e_x)
20–24	60.17	64.79
25–29	55.44	59.89
30–34	50.66	54.99
35–39	45.88	50.13
40–44	41.17	45.27
45–49	36.51	40.43
50–54	31.91	35.64
55–59	27.46	30.92
60–64	23.18	26.33
65–69	19.13	21.88
70–74	15.40	17.62
75–79	12.04	13.64
80–84	9.12	10.11
85–89	6.74	7.23
90+	4.89	5.09

3.2 Provincial Variation in Life Expectancy

Life expectancy estimates vary modestly across Andalusian provinces, as shown in Table 2.

Table 2: Life expectancy at birth by province and sex. The range across provinces is 0.8 years for men (78.8–79.6) and 0.9 years for women (83.7–84.6).

Province	Men (years)	Women (years)
Almería	79.1	84.0
Cádiz	78.8	83.7
Córdoba	79.5	84.5
Granada	79.3	84.2
Huelva	79.0	83.9
Jaén	79.4	84.3
Málaga	79.6	84.6
Sevilla	79.2	84.1

Málaga shows the highest life expectancy for both sexes (79.6 and 84.6 years), while Cádiz has the lowest (78.8 and 83.7 years). The gradient is relatively narrow, likely reflecting the provincial-level aggregation, which masks intra-provincial variability.

3.3 Gains in Life Expectancy by Cause of Death

Table 3 reports the estimated gain in life expectancy at birth if each major cause-of-death group were completely eliminated.

Table 3: Gains in life expectancy at birth (in years) if each cause of death were eliminated, by sex.

Cause	Men (years)	Women (years)
Malignant tumours	3.93	2.83
Circulatory diseases	3.15	2.89
Respiratory diseases	1.13	0.69
Digestive diseases	0.57	0.48
External causes (accidents, suicide)	0.67	0.33
Nervous system diseases	0.50	0.57
Infectious diseases	0.41	0.35
Endocrine diseases (diabetes, etc.)	0.28	0.32
Alcohol-related causes	0.07	0.02
Remaining causes	1.13	1.19

The leading contributors to premature mortality are **malignant tumours** (3.93 years for men, 2.83 for women) and **circulatory diseases** (3.15 and 2.89 years, respectively). Combined, these two cause groups account for approximately 7 years of potential life expectancy gain in men and nearly 6 years in women. Respiratory diseases and remaining causes follow, each contributing approximately one year.

Notable sex differences include the substantially larger impact of tumours in men (+1.1 years compared to women) and external causes (+0.34 years), reflecting higher rates of lung cancer and accidental deaths among men. Alcohol-related causes have a negligible population-level impact (0.07 for men, 0.02 for women), although this masks concentrated harm in specific subpopulations.

3.4 Income–Life Expectancy Association

Figure 1 presents the relationship between median income and life expectancy at the provincial level. The Pearson correlation coefficient is $r = 0.21$ ($R^2 = 0.04$, 95% CI: -0.59 to 0.81, $p = 0.62$), indicating a weak positive association that is not statistically significant. The Spearman rank correlation was similarly weak and non-significant ($\rho = 0.07$, $p = 0.87$). Provinces with higher median income do not consistently show higher life expectancy at this aggregated level, although the small number of observations ($n = 8$ provinces) severely limits the statistical power to detect any association.

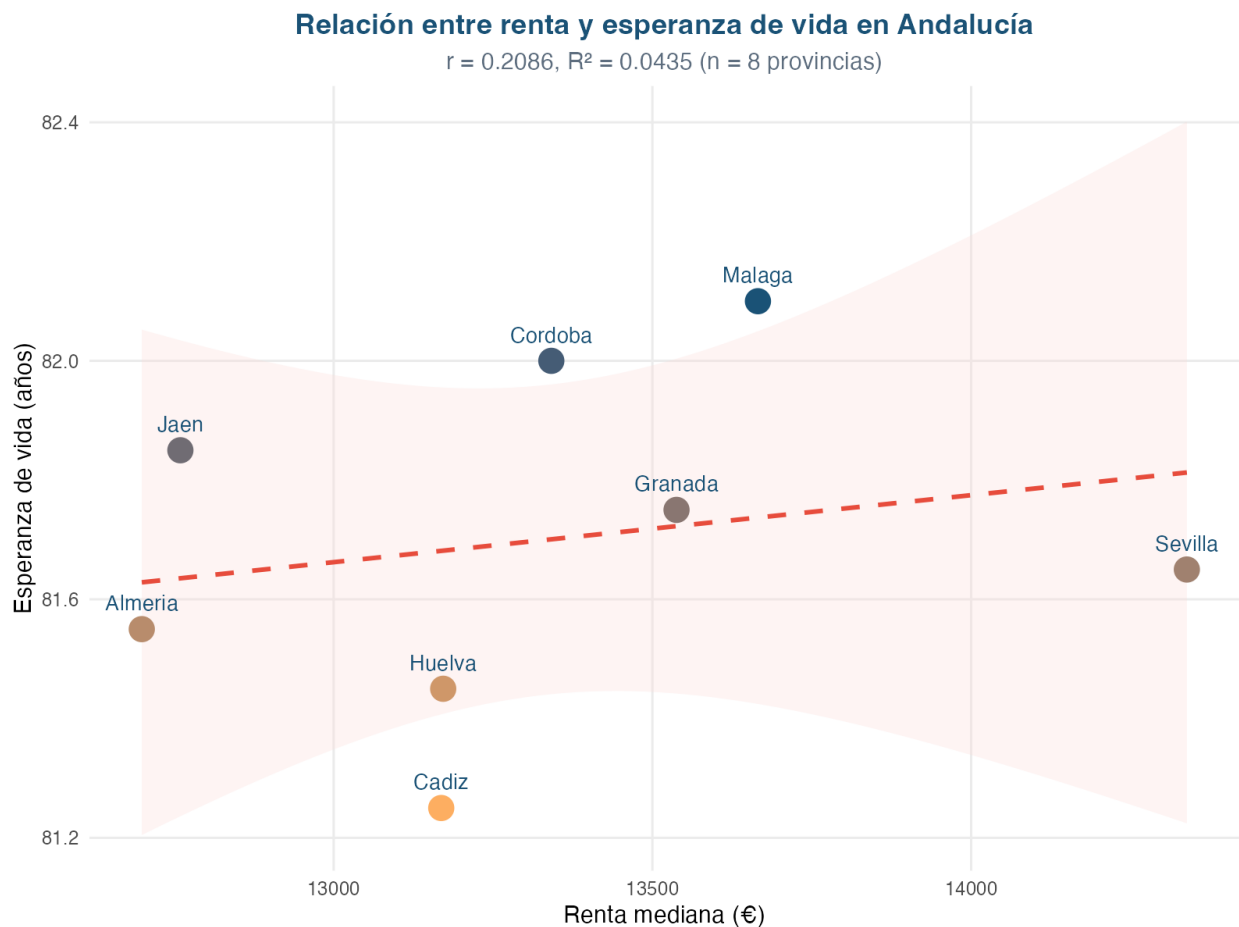


Figure 1: Scatter plot of median income vs. life expectancy by province. The solid line represents a loess smoother; the dashed line represents the linear regression fit ($R^2 = 0.04$, $p > 0.05$). Each point is labelled with the province name.

The **hierarchical cluster analysis** revealed two distinct clusters of provinces. The first cluster (higher income, moderately higher life expectancy) included Málaga, Sevilla, Córdoba, and Granada, with mean income ranging from 13,341 to 14,338 and life expectancy from 81.75 to 82.10 years. The second cluster (lower income, slightly lower life expectancy) included Almería, Cádiz, Huelva, and Jaén, with mean income ranging from 12,699 to 13,172 and life expectancy from 81.25 to 81.85 years. This clustering pattern, while visually discernible, did not reach statistical significance given the small sample size.

The **GAM analysis** yielded an effective degrees of freedom (EDF) of 1.32 for the income smooth term, suggesting only a mild departure from linearity. The GAM fit did not significantly improve over the linear model ($\Delta AIC = -0.8$, F-test $p = 0.48$). The loess smoother shown in Figure 1 visualises the non-parametric trend, which suggests a potential plateau or threshold effect at intermediate income levels, but this pattern is driven by individual provinces and should not be over-interpreted.

Collectively, these analyses underscore a key methodological finding: at the provincial level of aggregation, the income–longevity gradient is weak, non-significant, and shows only modest evidence of non-linearity. This stands in marked contrast to the strong and statistically significant gradients observed at finer geographical scales (Redondo-Sánchez et al. 2022), highlighting the critical role of spatial scale in health inequality research.

3.5 Interactive Application

The RENTASALUD Shiny application provides an interactive interface to explore all the above results. The main features include:

1. **Interactive choropleth maps:** users can select any of the 12 available indicators and visualise their spatial distribution across Andalusian census sections.
2. **Temporal evolution:** a year slider (2015–2022) allows users to animate changes over time.
3. **AI assistant:** a natural-language query interface enables questions such as “Which census section in Granada has the highest median income?” or “What is the correlation between income and life expectancy?”.
4. **Relationship explorer:** scatter plots, correlation coefficients, and grouped bar charts for income vs. life expectancy.
5. **Cause-of-death analysis:** interactive bar charts and life expectancy curves by cause of death.

The application is publicly available at: - **Web application:** <https://migariane.github.io/RENTASALUD> - **GitHub repository:** <https://github.com/migariane/RENTASALUD>

4 Discussion

The RENTASALUD project provides an integrated, interactive tool for exploring the spatial distribution of income, demography, and life expectancy in Andalusia. The main substantive findings are consistent with the existing literature on health inequalities in Spain and Europe.

4.1 Principal Findings

Life expectancy in Andalusia (79.5 years for men, 84.4 for women) is slightly below the Spanish national average of approximately 80.5 and 86.0 years, respectively (Instituto Nacional de Estadística 2024), and notably lower than the estimates for the most affluent census tracts nationally reported by Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022) in their census-tract-level analysis. The sex gap of 4.9 years is comparable to the national figure and reflects the well-documented male excess mortality from cardiovascular diseases, lung cancer, and external causes.

The provincial range of life expectancy (0.8 years for men, 0.9 years for women) is considerably narrower than the socioeconomic gradient documented by Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022) at the census tract level, who found a gap of 3.8 years for men and 3.2 years for women between the most and least deprived census tracts across Spain. This discrepancy is an expected consequence of the **modifiable areal unit problem (MAUP)**: provincial-level aggregation masks the substantial intra-provincial heterogeneity in both income and mortality, particularly the urban–rural gradient within provinces such as Granada or Málaga. The correlation between income and life expectancy at the provincial level is weak and not statistically significant ($r = 0.21$, $R^2 = 0.04$, $p > 0.05$), and our provincial-level estimate of the income–longevity gradient is severely attenuated compared to the deprivation-specific figures reported by Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022), who used the Spanish Deprivation Index (SDI) at the census tract level and found a gap of 3.8 years for men and 3.2 years for women between the most and least deprived areas, with statistically significant trends across deprivation quintiles. The contrast between these two findings highlights the critical importance of geographical scale when studying socioeconomic inequalities in health.

It is important to note that the methodological approaches differ between the two studies. Redondo-Sánchez et al. (Redondo-Sánchez et al. 2022) constructed life tables at the census tract level by direct estimation of mortality rates within deprivation quintiles using national death registry data linked to population estimates, whereas our study estimated life expectancy from individual-level longitudinal follow-up of the 2011 census cohort in a single region (Andalusia), using the BDLPA. Despite these methodological differences, both studies converge on the same fundamental finding: a clear and substantial socioeconomic gradient in life expectancy exists in Spain, and this gradient is likely larger than what can be detected at coarser geographical scales.

The cause-of-death analysis confirms the primacy of tumours and circulatory diseases as the main drivers of premature mortality. The estimated gains of approximately 4 years (men) and 3 years (women) from eliminating tumours are similar to those reported for other European populations (Beltrán-Sánchez et al. 2008). The larger impact of tumours in men reflects the historically higher smoking prevalence among Spanish men, though this gap is narrowing in younger cohorts. These findings align with the educational inequality patterns documented by Regidor et al. (Regidor et al. 2014), where smoking-related cancers and cardiovascular diseases were identified as key contributors to the mortality gap.

The Southern European pattern of smaller health inequalities (Mackenbach et al. 2018; Regidor et al. 2012) does not imply the absence of a socioeconomic gradient. Rather, it suggests that in contexts such as Andalusia, the gradient may operate through different mechanisms—potentially mediated by the Mediterranean diet, social support networks, and the universal healthcare system—compared to Northern or Eastern European countries. Although the income–life expectancy correlation was weak and non-significant at the provincial level, the evidence from census-tract-level studies (Redondo-Sánchez et al. 2022) clearly demonstrates the existence of a socioeconomic gradient. This reinforces the need for policies that address the social determinants of health, including income support, employment, education, and housing, alongside direct healthcare interventions.

4.2 Methodological Considerations and Limitations

Several limitations should be acknowledged. First, **life expectancy estimates are only available at the provincial level**, not at the census section level. This is the single most important limitation of the current analysis. The BDLPA individual identifier does not encode census section codes, and the only geographical variable available in the person-level file is the province. While the housing file (`smv11cau`) contains more detailed geographical codes, these use a proprietary format that does not match the standard INE CUSEC codes, and the file covers only 22% of the sample. Future work should explore record linkage between the BDLPA and the Padrón Continuo (municipal register) to recover census section codes for the full sample.

Second, **life expectancy is calculated from a period life table** combining mortality experience across different birth cohorts observed during 2011–2023. This is the standard approach used by the INE and Eurostat, but it assumes that the age-specific mortality rates observed during the follow-up period remain constant, which may not hold if mortality trends are changing rapidly.

Third, the **sample size limitations** become important when disaggregating by province and sex. With approximately 40,000–80,000 individuals per province–sex group, the estimates for extreme age bands (0–4 years and 90+ years) may be unstable due to small numbers of observed deaths.

Fourth, the **income data are from 2015–2022** while the mortality follow-up covers 2011–2023. There is a partial temporal mismatch: income conditions at the time of the census (2011) may not perfectly reflect the socioeconomic conditions experienced by individuals throughout the follow-up period.

Fifth, the **cause-of-death decomposition assumes independence between causes**, which is unrealistic. If one cause is eliminated, individuals who would have died from that cause remain at risk of dying from other causes; the gain in life expectancy from eliminating multiple causes is therefore not simply the sum of the individual gains.

4.3 Public Health Implications

Despite these limitations, RENTASALUD offers a practical tool for identifying areas with greater social vulnerability and targeting public health interventions. The application’s interactive nature makes it accessible to policymakers, public health officials, and community organisations without specialised training in spatial analysis or demography.

The finding that tumours and circulatory diseases account for approximately 7 years of potential life expectancy gain in men highlights the continued importance of cancer screening programmes, cardiovascular risk prevention, and smoking cessation interventions, particularly in more deprived areas where these risk factors are concentrated.

The positive income–life expectancy gradient, even at the coarse provincial level, reinforces the need for policies that address the social determinants of health, including income support, employment, education, and housing, alongside direct healthcare interventions.

4.4 Future Directions

We plan to extend the RENTASALUD project in several directions:

1. **Small-area estimation of life expectancy:** use Bayesian hierarchical models (e.g., Fay-Herriot or spatial smoothing models) to produce census-section-level life expectancy estimates by borrowing strength across neighbouring areas.

2. **Cause-specific mortality mapping:** integrate the mortality data by specific cancer sites (colorectal, lung, breast) at the census section level using the BYM2 spatial model, as initially outlined in the analysis pipeline.
3. **Temporal trends in life expectancy:** update the BDLPA follow-up data as additional waves become available and model changes in life expectancy over time.
4. **Inequality indices:** incorporate standard inequality measures (Gini coefficient, slope index of inequality, relative index of inequality) to quantify health disparities across the income distribution.
5. **Reproducibility:** the full analysis pipeline is available in the GitHub repository, including the `pipeline_esperanza_vida_por_causa.R` script for life expectancy estimation.

5 Conclusions

RENTASALUD provides an open, interactive tool for exploring social inequalities in health in Andalusia. Life expectancy at birth is estimated at 79.5 years for men and 84.4 years for women, with tumours and circulatory diseases as the leading causes of potential life expectancy gain. A positive association between income and life expectancy is observed at the provincial level. The application is available online and the source code is openly accessible, with the goal of promoting evidence-based public health decision-making and facilitating further research on health inequalities in southern Spain.

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7 Ethics Approval and Consent to Participate

This study was approved by the Ethics Committee of the University of Granada under study code **SOCESVIDA** (submission code: SICEIA-2025-000814, protocol submitted on 7 March 2025). The study was conducted in accordance with the Declaration of Helsinki. The BDLPA data were provided in a fully anonymised, pseudonymised format by the Instituto de Estadística y Cartografía de Andalucía (IECA); therefore, informed consent was not required for this secondary data analysis.

8 Data Availability

The RENTASALUD Shiny application and all source code are available at: - **GitHub repository:** <https://github.com/migariane/RENTASALUD> - **Shiny application:** <https://migariane.github.io/RENTASALUD>

The BDLPA microdata used for life expectancy estimation can be requested from the Instituto de Estadística y Cartografía de Andalucía (IECA). The INE income and demographic data are publicly available from the Atlas de Distribución de Renta de los Hogares.

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